FENOLLAR

Gidroksil guruhi benzol halqasiga bog‘langan birikmalar

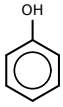
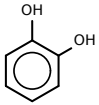
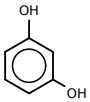
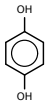
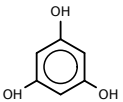
Student book – nazariy qo‘llanma

Molekulasida **gidroksil guruhi (–OH) benzol halqasidagi uglerod atomiga bevosita bog‘langan** organik birikmalar **fenollar** deyiladi. Eng oddiy vakili – **fenol C_6H_5OH** (karbol kislota). Bir atomli (mono-) fenollarning umumiy formulasi **$C_nH_{2n-6}O$** ($n \geq 6$); $M_r = 14n + 10$.

Muhim farq: –OH halqaga bog‘langan bo‘lsa – **fenol**, yon zanjirdagi uglerodga bog‘langan bo‘lsa (C_6H_5–CH_2OH) – **aromatik spirt**. Ularning xossalari keskin farq qiladi.

1. Tasnifi va nomlanishi

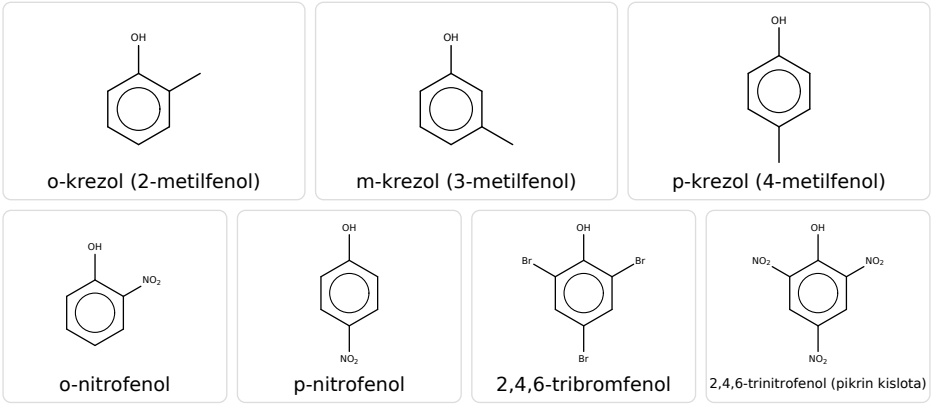
A) Gidroksil guruhlari soni bo‘yicha

Formula	Skelet	Sistematik nomi	Tarixiy nomi
C_6H_5OH		fenol (gidroksibenzol)	karbol kislota
$C_6H_4(OH)_2$ (1,2-)		benzol-1,2-diol	pirokatexin
$C_6H_4(OH)_2$ (1,3-)		benzol-1,3-diol	rezortsin
$C_6H_4(OH)_2$ (1,4-)		benzol-1,4-diol	gidroxinon
$C_6H_3(OH)_3$ (1,2,3-)		benzol-1,2,3-triol	pirogallol
$C_6H_3(OH)_3$ (1,3,5-)		benzol-1,3,5-triol	floroglyutsin

Bir atomli: fenol, krezollar; ikki atomli: pirokatexin, rezortsin, gidroxinon; uch atomli: pirogallol, floroglyutsin.

B) Halqadagi o‘rinbosarlarga ko‘ra nomlash

- Asos qilib **fenol** olinadi; gidroksil bog‘langan uglerod **1-raqamni** oladi.
- Ikki o‘rinbosarda **o-** (1,2-), **m-** (1,3-), **p-** (1,4-) qo‘shimchalari ishlatiladi.
- Metilfenollar – **krezollar** (C_7H_8O), dimetilfenollar – **ksilenollar**.



2. Tuzilishi

- Halqadagi 6 ta uglerod atomi **sp^2-gibridlangan**, molekula yassi.
- Kislorodning **juft elektronlari benzol halqasining π-sistemasi bilan qoplanadi** (p,π-qo‘shilish): elektron zichligi kislorodan halqaga siljiydi.
- Natijada: 1) **O–H bog‘i qutbliroq** bo‘ladi, vodorod oson ajraladi – fenol spirtdan **kuchliroq kislota**; 2) halqada **orto-** va **para-** holatlarda elektron zichligi oshadi – halqa benzoldan **faolroq**, o‘rin olish osonlashadi.
- Fenolda gidroksil va halqa bir-biriga ta’sir qiladi (atomlarning o‘zaro ta’siri – A.M. Butlerov nazariyasi).

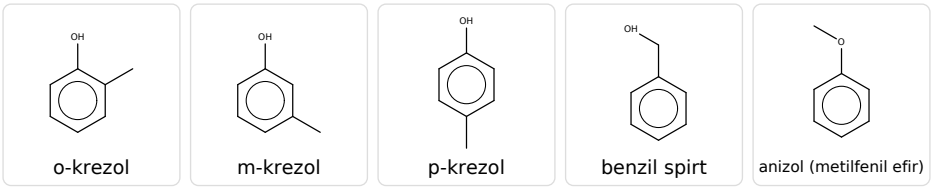
Modda	Formula	Kislotalik
etanol	C_2H_5OH	juda kuchsiz (suvdan kuchsizroq)
suv	H_2O	kuchsiz
fenol	C_6H_5OH	suvdan kuchli, H_2CO_3 dan kuchsiz
karbonat kislota	H_2CO_3	kuchsiz
sirka kislota	CH_3COOH	fenoldan kuchli

Kislotalik qatori: $C_2H_5OH < H_2O < C_6H_5OH < H_2CO_3 < CH_3COOH$

3. Izomeriyasi

Fenollarda **o‘rinbosarlarning halqadagi holati**, yon zanjir tuzilishi va **sinflararo** izomeriya uchraydi.

C_7H_8O tarkibli benzol halqali birikmalar – **5 ta** izomer: 3 ta krezol, aromatik spirt va oddiy efir:



$C_8H_{10}O$ tarkibli fenollar: 6 ta ksilenol va 3 ta etilfenol (jami 9 ta fenol).

4. Fizik xossalari

- Fenol – **rangsiz kristall modda**, t_{suyuq} = 41 °C, t_{qayn} = 182 °C, o‘ziga xos o‘tkir hidli.
- Havoda oksidlanib **pushti rangga** kiradi.
- Sovuq suvda **kam eriydi**; 66 °C dan yuqorida suv bilan istalgan nisbatda aralashadi. Spirt, efir va ishqorlarda yaxshi eriydi.
- Molekulalar orasida **vodorod bog‘lar** bor (qaynash harorati aren va alkandan yuqori).
- Suvdagi 5 % li eritmasi – **karbol kislota** (antiseptik).

Ehtiyot bo‘ling: fenol **zaharli** va teriga tushsa **kuydiradi**. Bug‘i nafas yo‘llarini zararlaydi; u bilan faqat qo‘lqop va shamollatish ostida ishlanadi.

5. Olinishi

1) Toshko‘mir smolasidan (koks gazi)

Smolani haydab, fenol fraksiyasi ajratiladi va ishqor bilan ishlanadi (natriy fenolyat), so‘ng kislota bilan fenol ajratiladi.